Practical No. 5

Real-Time Fraud Detection Using Streaming Financial Transactions

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Aim

To design and implement a real-time fraud detection system using streaming financial transaction data.

Learning Outcomes

After completing this experiment, the student will be able to:
- Understand real-time data streaming concepts
- Use Apache Kafka for transaction streaming
- Process streaming data using Spark Structured Streaming
- Detect fraudulent transactions in real time

Software & Hardware Requirements

Hardware:
- Minimum 8 GB RAM

Software:
- Ubuntu Linux / Windows
- Apache Kafka
- Apache Spark
- Python 3.x
- Java JDK 8 or higher

Dataset Link

Credit Card Fraud Detection — Kaggle (imbalanced transaction dataset):
https://www.kaggle.com/mlg-ulb/creditcardfraud/

Theory

Real-time fraud detection systems analyze transaction data as it is generated to identify suspicious patterns instantly.
Streaming frameworks like **Apache Kafka** and **Spark Structured Streaming** enable low-latency processing, making them suitable for financial fraud prevention.

System Architecture

Transaction Generator → Kafka Topic → Spark Streaming
↓
Fraud Detection Model
↓
Fraud Alerts / Logs

Procedure

Step 1: Create Kafka Topic

```
kafka-topics.sh --create --topic transactions --bootstrap-server
localhost:9092 --partitions 1 --replication-factor 1
```

Step 2: Start Kafka Producer (Transaction Stream)
Simulated financial transactions are streamed to Kafka.

Step 3: Start Spark Streaming Application
Spark consumes transaction data from Kafka.

Step 4: Apply Fraud Detection Rules
Transactions with unusually high amounts are flagged as fraud.

Step 5: Display Fraud Alerts
Fraudulent transactions are displayed on the console.

Output

- Live transaction stream
- Fraud alerts for suspicious transactions

Result

A real-time fraud detection system was successfully implemented using streaming data.

Applications

- Banking systems
- Online payment platforms
- Credit card fraud detection

Advantages

- Low-latency fraud detection
- Scalable streaming architecture
- Real-time decision making

Precautions

- Ensure Kafka and Spark services are running
- Handle data serialization properly

Conclusion

This experiment demonstrated how streaming frameworks can detect fraudulent financial transactions in real time.

Viva Voce Questions

1. What is real-time data streaming?
2. Why is Kafka used in fraud detection systems?
3. What are the advantages of Spark Streaming?